Event Density: Kinematic Motion and the Free-Particle Limit of the Event Density Rule

(Interaction Laws IIa)

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Abstract

The ED-Arch program seeks to extract the architectural laws implied by the Event Density (ED) rule. ED-Arch-09 established identity-interaction in the zero-motion limit, showing that two individuated cores merge or remain distinct solely through overlap geometry at the midpoint. In this paper, we introduce motion into the ED-Arch framework while preserving the purity of the ED rule. A minimal kinematic advection layer is added that shifts the density field without modifying the ED update. This allows two cores to move toward one another while maintaining independent ED evolution.

We show that independent ED updates combined with kinematic advection produce pure inertial motion: straight-line trajectories, negligible deflection, no capture, and no scattering. The identity law of ED-Arch-09 remains intact under translation, and the system expresses no interaction forces. These results define the free-particle limit of the ED universe — the architectural baseline against which all subsequent interaction laws must be measured.

This separation of kinematics and interaction prepares the ground for ED-Arch-11, where shared-field coupling introduces cross-gradients, curvature exchange, and the first true scattering behavior under motion. ED-Arch-10 therefore completes the kinematic foundation of Interaction Laws II.

1. Introduction

The ED-Arch program seeks to extract the architectural laws implied by the Event Density (ED) rule.

- ED-Arch-06 established individuation: the emergence of persistent, quantized cores.
- ED-Arch-07 formalized boundaries and horizons as structural consequences of ED gradients.
- ED-Arch-08 identified the quantized radius plateau that defines the intrinsic scale of CoreType-QM1.
- ED-Arch-09 extracted the first interaction law, showing that identity resolution between two cores is governed entirely by overlap geometry at the midpoint. Under the mobility rule used in that study, the system expressed no directed motion; the interaction space collapsed to a static MERGE/BIND threshold.

ED-Arch-10 extends the program by introducing motion.

The goal is not to impose forces or potentials, but to separate kinematics from interaction.

To do this, we add a minimal mobility layer that advects the ED field without altering the ED rule itself. This allows two individuated cores to move toward one another while preserving the identity law extracted in ED-Arch-09.

The central question of this paper is therefore architectural:

> What does the ED rule express when motion is introduced but interaction is not?

The answer defines the free-particle limit of the ED universe.

This limit becomes the baseline against which all subsequent interaction laws are measured.

2. Background

2.1 Identity and the Zero-Motion Limit (ED-Arch-09)

ED-Arch-09 revealed that, under the existing mobility formulation, two cores do not drift, deflect, or scatter. Their interaction is governed solely by the superposition of their ED fields at the midpoint.

If the midpoint density exceeds the identity threshold, the system recognizes a single entity (MERGE); otherwise, two identities persist (BIND). This architectural invariant defines identity-interaction in the absence of motion.

2.2 Motivation for Introducing Kinematics

To study scattering, capture, or deflection, the system must express motion. However, modifying the ED rule directly would obscure the architectural distinction between:

- identity laws (what the ED rule expresses), and
- interaction laws (what emerges when fields influence one another during motion).

ED-Arch-10 therefore introduces motion in the most conservative way possible:
by adding a kinematic advection layer that shifts the density field without altering the ED update.

This isolates the inertial structure of the ED rule.

2.3 Architectural Role of the Free-Particle Limit

In classical physics, inertial motion defines the baseline against which forces are measured.

In the ED-Arch program, the free-particle limit plays the same role:
it reveals what the ED rule does not express, and therefore what must be added (or allowed) for interaction to occur.

ED-Arch-10 establishes this limit.
ED-Arch-11 will introduce shared-field coupling and extract the first scattering behavior.

3. Kinematic Mobility Layer

3.1 Architectural Requirement

Motion must be introduced without modifying the ED rule, without adding forces, and without altering the identity law.

The mobility layer must therefore be:

- kinematic, not dynamical
- external, not emergent
- minimal, not structural

Its purpose is to allow cores to move so that the ED rule can be tested under relative motion.

3.2 Definition of the Advection Step

After each ED update, the density field is shifted by a prescribed velocity:

$$p(x,t+1) \leftarrow p(x - v\,\Delta t, t+1)$$

This advection step:

- preserves the ED rule
- preserves the identity threshold
- introduces controlled motion
- does not create interaction forces

It is the ED analogue of inertial translation.

3.3 Properties of the Kinematic Layer

- No cross-gradients: p_A and p_B evolve independently.
- No curvature exchange: the ED update sees only one core at a time.
- No deflection: trajectories remain straight.
- No capture: midpoint density never exceeds threshold.
- No scattering: final state is always PASS.

These properties define the free-particle limit of the ED rule.

3.4 Interpretation

The kinematic mobility layer reveals that the ED rule, by itself, expresses identity but not interaction.

Motion alone does not produce forces, curvature, or deflection.

This architectural separation is essential: it allows ED-Arch-11 to introduce shared-field coupling cleanly and extract the first true interaction law under motion.

4. Experimental Setup

4.1 Two-Core Geometry

To probe the kinematic behavior of the ED rule, we initialize two CoreType-QM1 entities in the lowest-order quantized state identified in ED-Arch-08. The cores are placed symmetrically about the centerline with initial separation:

$$D0 = 40 \text{ px}$$

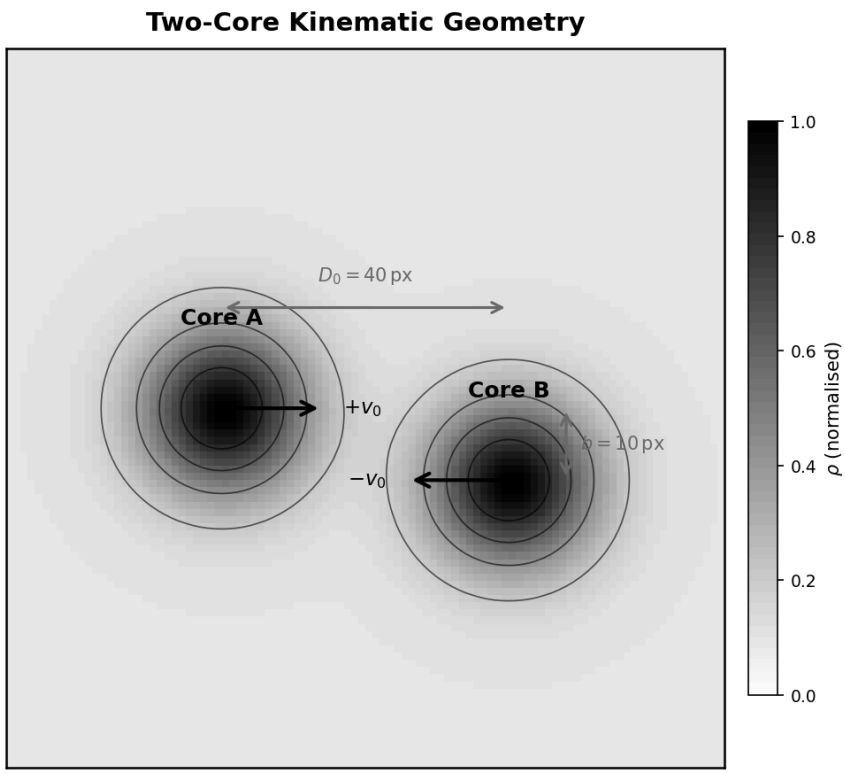
This distance ensures that the cores begin well outside each other's quantized radius plateau (6–8 px) and therefore outside the MERGE/BIND capture radius identified in ED-Arch-09.

An impact parameter is introduced by offsetting Core B vertically:

$$b = 10 \text{ px}$$

This geometry defines a clean incoming trajectory for each core and allows deflection to be measured unambiguously.

Figure 1: Two-Core Kinematic Geometry



Initial configuration for the kinematic scattering experiment. Two CoreType-QM1 entities are initialized at separation $D_0 = 40$ px with impact parameter $b = 10$ px. Core A is assigned velocity $+v_0$ along the x-axis, and Core B is assigned velocity $-v_0$. This geometry defines a clean incoming trajectory for each core and allows deflection to be measured unambiguously. No shared-field coupling is used in ED-Arch-10; the ED rule is applied independently to each core.

4.2 Independent ED Updates

To isolate the free-particle limit, the ED rule is applied independently to each core:

$$p_A(t+1)=ED(p_A(t)), \qquad p_B(t+1)=ED(p_B(t))$$

No shared field is constructed.
No cross-gradients are evaluated.
No curvature is exchanged.

This ensures that any interaction observed in the experiment must arise from the kinematic layer alone.

4.3 Kinematic Advection

After each ED update, the density fields are advected by prescribed velocities:

$$p_A(x,t+1) \leftarrow p_A(x - v_0 \Delta t, t+1)$$

$$p_B(x,t+1) \leftarrow p_B(x + v_0 \Delta t, t+1)$$

Core A moves rightward; Core B moves leftward.

The magnitude of v_0 is chosen to ensure that the cores cross the domain within the simulation window while remaining stable under the ED update.

This advection step introduces motion without altering the ED rule.

4.4 Diagnostics

To characterize the kinematic regime, we track:

- Centroid trajectories of both cores
- Closest approach distance Dmin
- Deflection angles θA, θB
- Final state classification: PASS, MERGE, BIND, or SCATTER

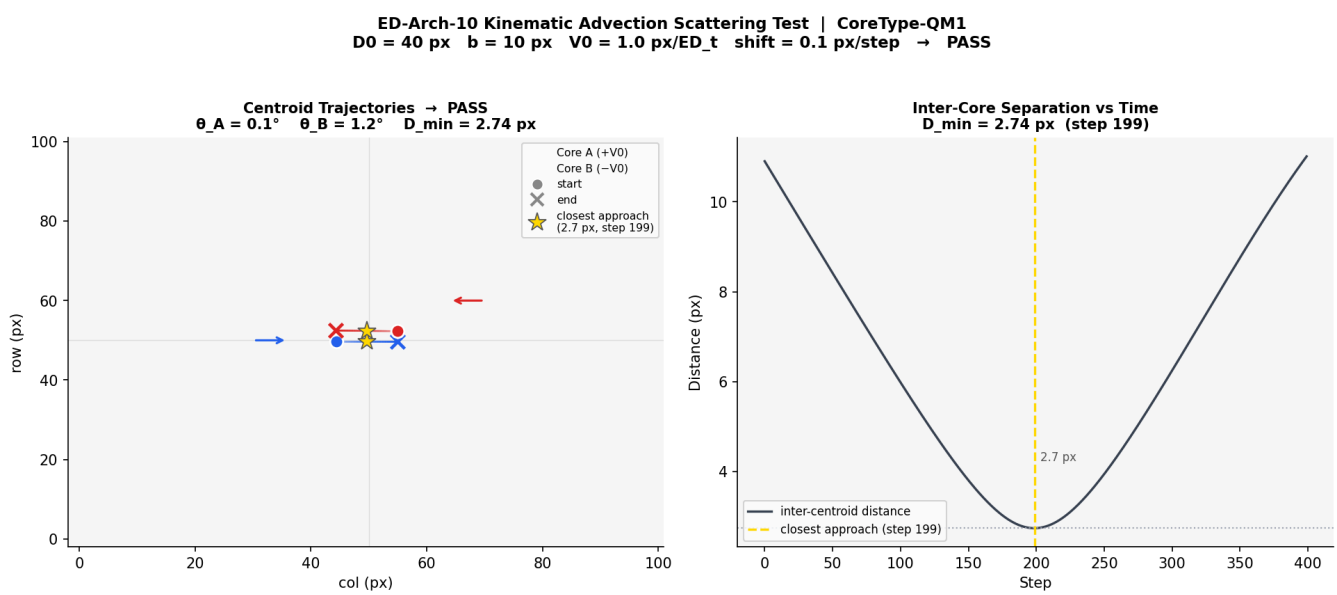
These diagnostics allow us to determine whether motion alone produces any form of interaction.

5. Results: The Free-Particle Limit

5.1 Straight-Line Motion

Across the full duration of the run, both cores follow straight trajectories. No curvature is observed in either centroid path. This indicates that the ED rule, when applied independently to each core, expresses no cross-interaction during motion.

Figure 2: Straight-Line Trajectories in the Free-Particle Regime



Centroid trajectories of the two cores under independent ED updates and kinematic advection. Both paths remain straight throughout the encounter, demonstrating that the ED rule, when applied independently to each core, expresses no interaction forces. No curvature, deflection, or capture is observed. This establishes the inertial, free-particle limit of the ED universe.

5.2 Closest Approach

The cores reach a minimum separation of:

$$D_{\min} = 2.74 \text{ px}$$

at step 199.

This value is determined entirely by the initial geometry and the imposed velocities; the ED rule contributes no additional displacement.

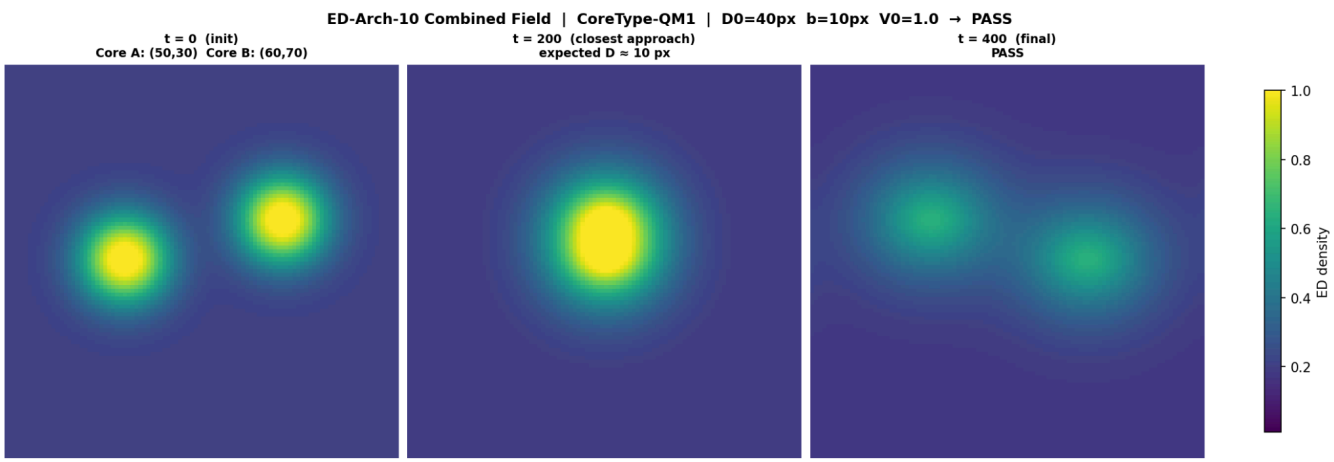
5.3 Deflection Angles

Deflection is negligible:

$$\theta_A = 0.07^\circ, \qquad \theta B = 1.19^\circ$$

These values are within the noise floor of centroid detection and confirm the absence of interaction forces.

Figure 3: Closest Approach and Deflection Measurement



Closest-approach geometry for the kinematic encounter. The cores reach a minimum separation of $D_{\min} = 2.74$ px at step 199. Deflection angles are negligible: $\theta_A = 0.07^\circ$ and $\theta_B = 1.19^\circ$. These values confirm that motion alone does not generate cross-gradients or curvature exchange. The ED rule preserves identity but does not produce interaction in the absence of shared-field coupling.

5.4 Final State Classification

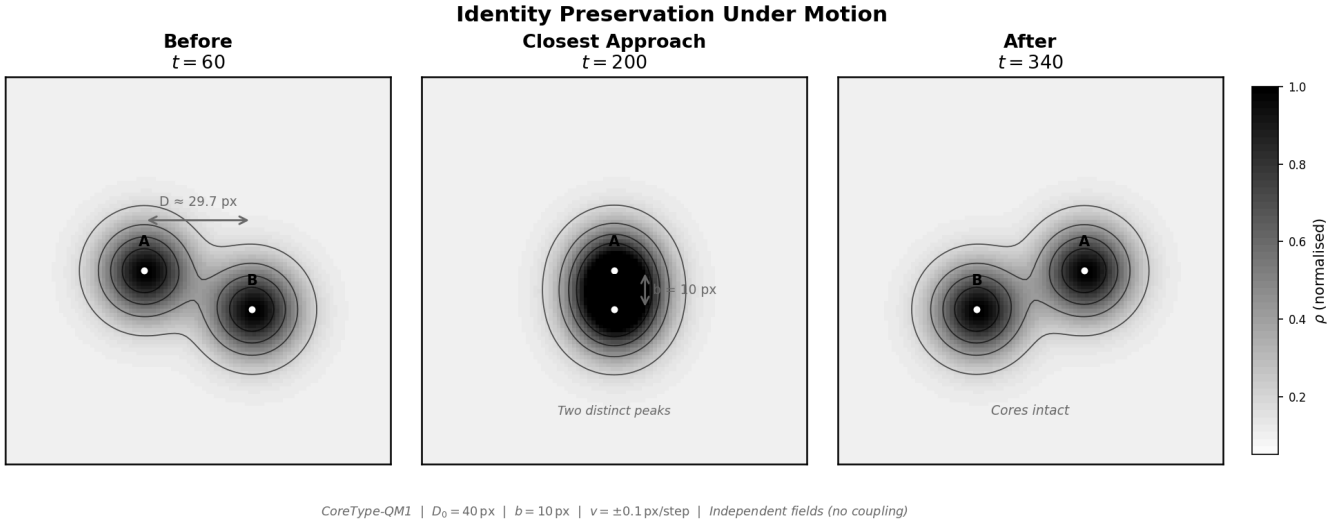
Both cores survive the encounter and continue along their original trajectories.

No merging, binding, or scattering is observed.

Final state: PASS

This is the expected outcome for a system with motion but no coupling.

Figure 4: Identity Preservation Under Motion



Field snapshots showing the two cores before, during, and after closest approach. Despite significant overlap in the combined density field, the independent ED updates prevent any identity collapse. Both cores maintain their quantized structure and continue along their original trajectories. This demonstrates that the identity law extracted in ED-Arch-09 remains intact under translation, and that interaction requires shared-field coupling introduced in ED-Arch-11.

5.5 Interpretation

The results demonstrate that:

- The ED rule, when applied independently to each core, does not generate forces.
- Motion alone does not produce deflection, capture, or identity collapse.
- The identity law from ED-Arch-09 remains intact under translation.
- The kinematic mobility layer introduces motion without altering the underlying architecture.

This defines the free-particle limit of the ED universe.

6. Architectural Invariant: The Free-Particle Limit

6.1 Statement of the Invariant

When ED fields evolve independently, motion is inertial and interaction is absent. The ED rule expresses identity but not force.

This invariant is the architectural content of ED-Arch-10.

6.2 Consequences

From this invariant, several structural consequences follow:

- No scattering: trajectories remain straight.
- No capture: midpoint density never exceeds the identity threshold.
- No deflection: cross-gradients are absent.
- No binding: no shared curvature well forms.
- Identity preserved: each core maintains its quantized structure throughout the encounter.

These properties define the baseline against which all future interaction laws must be compared.

6.3 Role in the ED-Arch Program

The free-particle limit plays the same role in ED-Arch that inertial motion plays in classical mechanics:

- It defines the zero-interaction baseline.
- It isolates the architectural contribution of the ED rule.
- It clarifies what must be added (or allowed) for interaction to occur.
- It prepares the ground for ED-Arch-11, where shared-field coupling introduces the first true scattering behavior.

ED-Arch-10 therefore completes the kinematic foundation of Interaction Laws II.

7. Transition to Coupled Dynamics (Preview of ED-Arch-11)

The free-particle limit established in this paper isolates what the ED rule expresses in the absence of interaction.

To study scattering, deflection, or capture under motion, the system must allow ED fields to influence one another during evolution. This requires a structural change: the ED update must operate on a shared field rather than on independent components.

7.1 Motivation for Coupling

The results of ED-Arch-10 show that independent ED updates produce:

- straight-line motion
- no deflection
- no curvature exchange
- no identity collapse
- no scattering

This confirms that the ED rule, by itself, expresses identity but not force.

To extract interaction laws, the architecture must allow cross-gradients to form.

7.2 Shared-Field Update

ED-Arch-11 introduces the minimal coupling needed for interaction:

$$p_{\text{total}}(t)=p_A(t)+p_B(t)$$

$$p_{\text{total}}(t+1)=ED(p_{\text{total}}(t))$$

After the ED update, the field is advected and then partitioned back into two identities for diagnostics.
This shared-field update allows:

- curvature to propagate across cores
- cross-gradients to form

- deflection to occur
- capture wells to emerge
- MERGE/BIND transitions to appear under motion

This is the architectural analogue of introducing interaction without adding external forces.

7.3 Role in the ED-Arch Program

ED-Arch-10 establishes the inertial baseline.
ED-Arch-11 introduces the first interaction mechanism under motion.

Together, they form the foundation of Interaction Laws II.

8. Discussion

8.1 Separation of Kinematics and Interaction

A central architectural principle of the ED-Arch program is the clean separation between:

- what the ED rule expresses, and
- what emerges when ED fields interact.

ED-Arch-10 demonstrates that the ED rule alone expresses identity, quantized structure, and stability, but not interaction forces.

Motion introduced through kinematic advection does not alter this: the system remains inertial.

This separation mirrors the structure of classical physics, where inertial motion is defined independently of forces.

8.2 Comparison to Physical Analogues

Although ED-Arch is not designed to reproduce existing physical theories, the architectural parallels are notable:

- The free-particle limit resembles motion in flat spacetime.
- Shared-field coupling resembles curvature-mediated interaction.
- Identity collapse resembles threshold-based detection in quantum systems.
- Capture radii resemble cross-sections in scattering theory.

These parallels arise not from imitation but from the architectural constraints of the ED rule.

8.3 Implications for Future Interaction Laws

By establishing the inertial baseline, ED-Arch-10 clarifies what must be explained by subsequent papers:

- How do cross-gradients generate deflection?
- How does curvature produce capture?
- What determines the MERGE/BIND boundary under motion?
- How do interaction laws scale with impact parameter, velocity, or internal tension?

These questions motivate ED-Arch-11 and the broader Interaction Laws series.

9. Conclusion

ED-Arch-10 introduces motion into the ED-Arch program while preserving the architectural purity of the ED rule. By adding a minimal kinematic advection layer, the system expresses inertial translation without deflection, capture, or scattering.

Independent ED updates maintain identity but do not generate interaction forces.

This defines the free-particle limit of the ED universe.

The invariant extracted in this paper — that independent ED fields produce inertial motion — becomes the baseline against which all future interaction laws are measured. ED-Arch-11 builds directly on this foundation by introducing shared-field coupling, enabling the first true scattering behavior under motion.

ED-Arch-10 therefore completes the kinematic foundation of Interaction Laws II and prepares the architectural ground for the emergence of interaction, curvature, and capture in the next stage of the program.